RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2018- 2019 (Odd Semester)

<u>Innovative Practices Description</u>

Unit-II Image Enhancement

Degree, Semester & Branch: VII Semester B.E. ECE A & B
Course Code & Title: IT6005 Digital image Processing
Name of the Faculty member: Mr.K.Ragavan
Name of the Topic: Spatial and Frequency domain Filtering
Name of the Innovative Practice: MATLAB Simulation
ICT Tool Used: MATLAB Software and LCD Projector

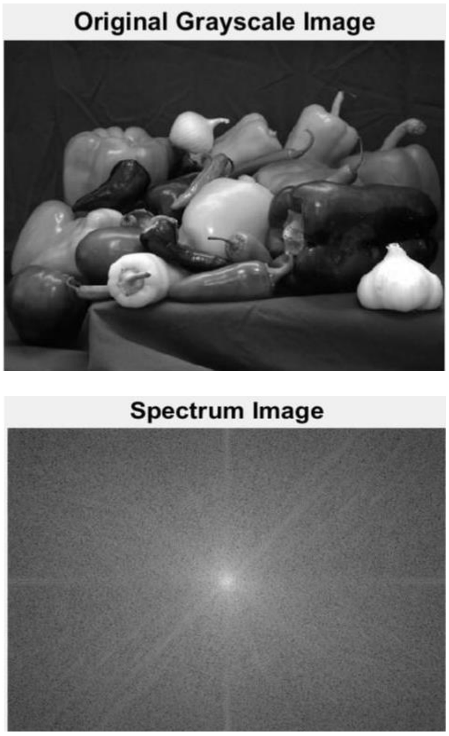
Description:

MATLAB is one of the most popular fourth-generation programming languages in the world. It is one of the best numerical analysis environments. MATLAB is also the most powerful and high-performance language that is used in technical computing. It is built to solve the problems which use mathematics notations. Use of black-boards or the typical lecture methods are not adequate to teach science and other related subjects. The evolution of a process can be conveyed better with simulation. It helps to promote creativity and motivate the students to do projects using MATLAB Software.

```
grayImage = imread('peppers.png');
% Get the dimensions of the image.
% numberOfColorBands should be = 1.
[rows, columns, numberOfColorBands] = size(grayImage);
if numberOfColorBands > 1
  % It's not really gray scale like we expected - it's color.
  % Convert it to gray scale by taking only the green channel.
  grayImage = grayImage(:, :, 2); % Take green channel.
end
% Display the original gray scale image.
subplot(2, 1, 1);
imshow(grayImage, []);
fontSize = 20;
title('Original Grayscale Image', 'FontSize', fontSize, 'Interpreter', 'None');

% Set up figure properties:
% Enlarge figure to full screen.
set(gcf, 'Units', 'Normalized', 'OuterPosition', [0 0 1 1]);
% Get rid of tool bar and pulldown menus that are along top of figure.
set(gcf, 'Toolbar', 'none', 'Menu', 'none');
% Give a name to the title bar.
set(gcf, 'Name', 'Demo by ImageAnalyst', 'NumberTitle', 'Off')

% Display the original gray scale image.
subplot(2, 1, 2);
F=fft2(grayImage);
S=fftshift(log(1+abs(F)));
imshow(S,[]);
title('Spectrum Image', 'FontSize', fontSize, 'Interpreter', 'None');
```



Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	3	3	2	2			1		1	2	3

Reference:

- https://in.mathworks.com/matlabcentral/answers/261707-how-to-convert-an-image-to-frequency-domain-in-matlab
- Rafael C. Gonzalez, Richard E. Woods, Steven L. Eddins, "Digital Image Processing Using MATLAB", Third Edition Tata Mc Graw Hill Pvt. Ltd., 2011.

CO2: The students will be able to examine various types of images, intensity transformations and spatial filtering and to develop Fourier transform for image processing in frequency domain.